No. of Printed Pages : 5

Sem-II-UG-CompSc-MJP-4(NEP)
(DBMS)

DT-06/08/2025

2024-25

Time - 3 hours

Full Marks - 100

Answer all parts as per instructions.
Figures in the right hand margin indicate marks.

PART – I

1. Answer all questions by choosing the correct answer from the given alternatives. [1 × 10

(a) Roll back of transactions is normally used to

(i) Retrieve old records

(ii) Update the transaction

(iii) Recover from transaction failure

(iv) Repeat a transaction

(b) The ability to query data, as well as insert, delete and alter tuples, is offered by ________.

(i) DCL (ii) DDL

(iii) DML (iv) TCL

(c) WHERE clause is used to filter ________.

(i) Rows

P.T.O.

(ii) Groups

(iii) Columns

(iv) Tables

(d) Which one of the following refers to the total view of the database content ?

(i) Conceptual view (ii) Physical view

(iii) Internal view (iv) External view

(e) __________ means that the data used during the execution of a transaction cannot be used by a second transaction until the first one is completed.

(i) Consistency (ii) Atomicity

(iii) Durability (iv) Isolation

(f) In which normal form, every determinant in a table must be a candidate key ?

(i) First (ii) Second

(iii) Boyce-Codd (iv) Fifth

(g) Select the correct foreign key constraint ?

(i) Entity Integrity (ii) Referential integrity

(iii) Domain integrity (iv) None of these

(h) An __________ is a set of entities of the same type that shares the same properties or attributes.

(i) Entity set (ii) Attribute set

(iii) Relation set (iv) Entity model

(i) Which of the following command is used to create a database in SQL ?

(i) MAKE (ii) CREATE

(iii) INSERT (iv) DEVELOP

(j) Relational database schema normalization is NOT for

(i) Eliminating uncontrolled redundancy of data stored in the database

(ii) Eliminating number of anomalies that could otherwise occur with inserts and deletes

(iii) Ensuring that functional dependencies are enforced

(iv) Reducing the no. of joins required to satisfy a query

PART – II

2. Answer all questions within 50 words each. [2 × 9

(a) What is Transaction ?

(b) What is DML compiler ?

(c) What is a Foreign key ?

(d) What is an Entity type ?

(e) What is the difference between Unique key and Primary key ?

(f) What is Multi-valued dependency ?

(g) What is a JOIN in SQL ?

(h) What is 2NF ?

(i) What are the states of transaction ?

PART – III

3. Answer any eight of the following questions within 250 words each. [5 × 8

(a) Explain differences between entity and attributes, Entity set vs. Relationship set, Binary vs. Ternary relationship.

(b) What is Relational Algebra ? What are its properties ?

(c) What are the differences between Shared lock and Exclusive Lock ?

(d) Define Schedule and Serializability ? Explain types of Schedule.

(e) Explain functional dependency with an example ? What are trivial and non-trivial functional dependencies ?

(f) What do you mean by join ? Explain Inner-join with example.

(g) What is Data Model ? Explain Network model with the help of diagram.

(h). How to create a table in a database by using SQL ? What is the difference between HAVING and WHERE clauses ?

(i) Explain Armstrong axioms.

(j) Write different types of Keys. Explain all.

PART – IV

*Answer **any four** questions within 800 words each.*

4. What is Normalization ? Explain 1NF, 2NF, 3NF and BCNF with examples. [8

5. What is Data Independence ? Explain differences between Physical and Logical data independence ? [8

6. Write Fundamental operations in Relational Algebra and explain with examples. (selection, projection, Cartesian product, set operations, join operations) [8

7. Draw Transaction state diagram and describe each state that a transaction goes through during its execution. [8

8. Draw ER Diagram for : [4 × 2

(i) University management system

(ii) Online Shopping